THE COLLATZ CONJECTURE AS AN ORBIT-CLOSURE PROBLEM IN THE AFFINE GROUP $\mathrm{Aff}\big(\mathbb{Z}\big[\frac{1}{6}\big]\big)$

RESEARCH NOTES

Abstract. We reformulate the Collatz conjecture as an orbit-closure statement in the affine group $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$. Every word w in the two generators $\mathsf{L}\colon x \mapsto x/2$ and $\mathsf{R}\colon x \mapsto 3x+1$ acts as an affine map $f_w(x) = ax + b$ with slope $a = 3^r/2^\ell$ and intercept $b \in \mathbb{Z}[\frac{1}{6}]$. We prove that the Collatz semigroup $\mathcal{S} = \langle \mathsf{L}, \mathsf{R} \rangle$ embeds faithfully into $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$ (Theorem 3.2), characterise the word-equality relation (Theorem 3.3), and show the conjecture is equivalent to the backward orbit of 1 under $\mathcal{S}$ covering all of $\mathbb{N}$ (Theorem 4.3). We establish that $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$ is solvable and hence amenable (Proposition 5.1), identifying this as a structural advantage over Thompson-group approaches where amenability of the ambient group is itself an open problem. The residual gap between density-1 coverage and full coverage is identified precisely, and the conjecture is reduced to a concrete subset-sum reachability question over $\mathbb{Z}[\frac{1}{6}]$ (Question 6.1).

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1. Introduction

The Collatz conjecture asserts that iterating the map

$$T(n) \;=\; \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n+1 & \text{if } n \text{ is odd,} \end{cases}$$

from any positive integer n eventually reaches 1. Despite its elementary statement the problem has resisted all attempts at proof since Collatz posed it informally around 1937. The strongest partial result is due to Tao [11], who showed that the Collatz orbit of almost every positive integer (in the logarithmic density sense) reaches arbitrarily small values.

This paper develops a new algebraic framework centred on the *affine group* $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$ of transformations $x \mapsto ax+b$ over the ring $\mathbb{Z}[\frac{1}{6}] = \mathbb{Z}[\frac{1}{6}]$. The even and odd branches of T are each affine maps, every Collatz orbit prefix composes to a single affine map, and the Collatz conjecture becomes a question about orbit coverage under a semigroup action—a setting with clear algebraic structure.

Main results.

- **Faithful embedding** (Section 3). The Collatz semigroup $\mathcal{S} = \langle \mathsf{L}, \mathsf{R} \rangle$ embeds faithfully into $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$. Two words are equal as semigroup elements if and only if they have the same number of each generator *and* the same intercept. All $\binom{r+\ell}{r}$ orderings of a word with r copies of R and ℓ copies of L produce distinct affine maps. The word problem is decidable.
- **Orbit reformulation** (Section 4). The conjecture is equivalent to: for every $n \in \mathbb{N}$, there exists $f \in \mathcal{S}$ with $f(n) = 1$.
- **Amenability** (Section 5). $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$ is solvable of derived length 2, hence amenable by Day's theorem [4]. This gives density-1 coverage for free, recovering Tao's result in this language.
- **The residual gap** (Section 6). Amenability alone does not close the gap between density-1 and full coverage. The precise obstruction is reduced to a subset-sum reachability problem over $\mathbb{Z}[\frac{1}{6}]$.

Comparison with Thompson-group approaches. Several authors have explored embeddings of the Collatz dynamics into Thompson's groups F, T, and V [3, 2]. A key difficulty is that amenability of Thompson's group F is a major open problem [1, 5]. Any proof relying on amenability of the ambient group therefore cannot use F without first resolving that problem. The affine group $\mathrm{Aff}(\mathbb{Z}[\frac{1}{6}])$ is *provably* amenable, making the present framework structurally simpler.

The related group $G_{(2,3)}$ of piecewise-linear homeomorphisms with breakpoints in $\mathbb{Z}[\frac{1}{6}]$ and slopes in $\langle 2, 3 \rangle$ is discussed in Section 8 as a direction for more rigid embedding.

2. Setup and notation

Definition 2.1 (The ring $\mathbb{Z}[\frac{1}{6}]$)**.** Write $\mathbb{Z}[\frac{1}{6}] = \mathbb{Z}[\frac{1}{6}]$ for the smallest subring of $\mathbb{Q}$ containing $\mathbb{Z}$ and closed under division by 2 and by 3:

$$\mathbb{Z}[\tfrac{1}{6}] \;=\; \Big\{ \frac{p}{2^a 3^b} : p \in \mathbb{Z},\ a, b \geq 0 \Big\}.$$

The units of $\mathbb{Z}[\frac{1}{6}]$ are $\{\pm 2^s 3^t : s, t \in \mathbb{Z}\}$.

Definition 2.2 (Affine group)**.** The *orientation-preserving affine group* is

$$\mathrm{Aff}^+(\mathbb{Z}[\tfrac{1}{6}]) \;=\; \big\{x \mapsto ax+b : a = 2^s 3^t > 0,\ s,t \in \mathbb{Z},\ b \in \mathbb{Z}[\tfrac{1}{6}]\big\},$$

with composition $(a_1, b_1) \circ (a_2, b_2) = (a_1a_2,\, a_1b_2 + b_1)$, identity $(1, 0)$, and inverse $(a, b)^{-1} = (a^{-1}, -b/a)$. We write elements as pairs (a, b) and the corresponding map as $f_{(a,b)}$.

Definition 2.3 (Generators)**.** Define two elements of $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$:

$$\begin{aligned} \mathsf{L} &= \big(\tfrac{1}{2},\, 0\big), & f_{\mathsf{L}}(x) &= x/2 \quad \text{(even branch of } T\text{)},\\ \mathsf{R} &= (3,\, 1), & f_{\mathsf{R}}(x) &= 3x+1 \quad \text{(odd branch of } T\text{)}. \end{aligned}$$

Their group inverses are $\mathsf{L}^{-1} = (2, 0)$ (that is, $x \mapsto 2x$) and $\mathsf{R}^{-1} = (\frac{1}{3}, -\frac{1}{3})$ (that is, $x \mapsto (x-1)/3$).

Definition 2.4 (Collatz semigroup and words)**.** The *Collatz semigroup* is $\mathcal{S} = \langle \mathsf{L}, \mathsf{R}\rangle \subset \mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$. A *word* is a finite string $w = c_1c_2\cdots c_n$ with $c_i \in \{\mathsf{L}, \mathsf{R}\}$; its *type* is (r, ℓ) where r counts R's and ℓ counts L's. Composition is left-to-right: $f_w = f_{c_n} \circ \cdots \circ f_{c_1}$, so $f_w(x)$ is obtained by applying f_{c_1} first.

Remark 2.5 (Canonical word of an orbit)**.** The Collatz sequence $n, T(n), T^2(n), \ldots, 1$ determines a canonical word $w(n)$: write L for each even step and R for each odd step. Then $f_{w(n)}(n) = 1$ if and only if the sequence terminates at 1.

3. The affine embedding

3.1. Every word is an affine map.

Lemma 3.1 (Affine structure)**.** *For any word $w = c_1 \cdots c_n$ of type (r, ℓ), the element $f_w \in \mathcal{S}$ has the form $f_w(x) = a_w x + b_w$ where*

$$a_w = \frac{3^r}{2^\ell}, \tag{1}$$

$$b_w = \sum_{\substack{i=1\\ c_i=\mathsf{R}}}^{n} \frac{3^{r_i}}{2^{\ell_i}}, \tag{2}$$

and $r_i = \#\{j < i : c_j = \mathsf{R}\}$, $\ell_i = \#\{j < i : c_j = \mathsf{L}\}$ denote the number of R's (resp. L's) strictly before position i.

Proof. By induction on n. The base cases $f_{\mathsf{L}}(x) = \frac{1}{2}x + 0$ and $f_{\mathsf{R}}(x) = 3x + 1$ match (1)–(2).

Suppose $f_v(x) = ax + b$ for a word v of type (r', ℓ').
Appending L*:* $f_{v\mathsf{L}}(x) = f_v(x/2) = \frac{a}{2}x + \frac{b}{2}$. The new slope is $a/2 = 3^{r'}/2^{\ell'+1}$, consistent with (1) for type $(r', \ell' + 1)$. Appending L adds no new term to (2), consistent with the sum being over R-positions only.
Appending R*:* $f_{v\mathsf{R}}(x) = f_v(3x + 1) = 3ax + (a + b)$. The new slope is $3a = 3^{r'+1}/2^{\ell'}$, consistent with (1) for type $(r' + 1, \ell')$. The new intercept is $a + b = 3^{r'}/2^{\ell'} + b$, which is b plus the new term $3^{r'}/2^{\ell'}$ contributed by the appended R at position $n + 1$ (which has $r' = r'$ preceding R's and ℓ' preceding L's), consistent with (2). $\square$

3.2. **Injectivity.**

Theorem 3.2 (Faithful embedding)**.** *The map $\mathcal{S} \to \mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$, $w \mapsto (a_w, b_w)$, is injective. The Collatz semigroup is isomorphic to its image in $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$.*

Proof. Suppose $f_{w_1} = f_{w_2}$, i.e. $a_{w_1} = a_{w_2}$ and $b_{w_1} = b_{w_2}$. By (1) and unique factorisation in $\mathbb{Z}$, equality of slopes $3^{r_1}/2^{\ell_1} = 3^{r_2}/2^{\ell_2}$ forces $r_1 = r_2 = r$ and $\ell_1 = \ell_2 = \ell$.

Now suppose $w_1 \neq w_2$ are two distinct words of the same type (r, ℓ). Let k be the first position at which they differ. Without loss of generality, $c_k^{(1)} = \mathsf{R}$ and $c_k^{(2)} = \mathsf{L}$. Both words agree on positions $1, \ldots, k-1$, so the partial counts $(r_k^{(1)}, \ell_k^{(1)}) = (r_k^{(2)}, \ell_k^{(2)})$ are identical; call them (r_0, ℓ_0). The term $3^{r_0}/2^{\ell_0} \in \mathbb{Z}[\frac{1}{6}]$ appears in b_{w_1} (contributed by position k) but not yet in b_{w_2}. After position k, the remaining letters of w_1 and w_2 contribute terms with denominators divisible by a power of 2 at least $\ell_0 + 1$ (since w_2 has an L at position k, incrementing ℓ), while $3^{r_0}/2^{\ell_0}$ has denominator exactly 2^{ℓ_0}. Since elements of $\mathbb{Z}[\frac{1}{6}]$ with different denominators are distinct, and since the remaining contributions to b_{w_2} cannot cancel the term $3^{r_0}/2^{\ell_0}$ present only in b_{w_1}, we conclude $b_{w_1} \neq b_{w_2}$. □

3.3. **Word equality and the word problem.**

Theorem 3.3 (Word equality)**.** *Two words $w_1, w_2 \in \{\mathsf{L}, \mathsf{R}\}^*$ represent the same element of $\mathcal{S}$ if and only if:*

(i) *w_1 and w_2 have the same type (r, ℓ), and*
(ii) *$b_{w_1} = b_{w_2}$.*

All $\binom{r+\ell}{r}$ orderings of a type-(r, ℓ) word produce pairwise distinct affine maps. The word problem for $\mathcal{S}$ is decidable in time $O(|w|)$.

Proof. Conditions (i)–(ii) are necessary by Theorem 3.2. They are sufficient by definition of f_w. Distinctness of all $\binom{r+\ell}{r}$ orderings follows from the injectivity argument: any two distinct orderings of the same (r, ℓ)-type words differ at some first position, and the argument in the proof of Theorem 3.2 shows their intercepts differ. Decidability: given words w_1, w_2, compute (a_{w_i}, b_{w_i}) using Lemma 3.1 in $O(|w_i|)$ time, then compare. □

Example 3.4 (First non-trivial relation)**.** The words $\mathsf{LRRLRLL}$ and $\mathsf{RRLLLLR}$ both have type $(r, \ell) = (3, 4)$. By Lemma 3.1:

$$f_{\mathsf{LRRLRLL}}(x) = f_{\mathsf{RRLLLLR}}(x) = \frac{27}{16}\,x + \frac{7}{4}.$$

This is the shortest non-trivial relation in the semigroup. Both words produce the same intercept $b = \frac{7}{4}$, which can be verified by computing the contribution of each R-position.

Corollary 3.5 (Group structure)**.** *The semigroup $\mathcal{S}$ satisfies none of the standard relations that would identify it with a classical group:*

$$f_{\mathsf{LR}}(x) = \tfrac{3}{2}x + 1, \quad f_{\mathsf{RL}}(x) = \tfrac{3}{2}x + \tfrac{1}{2},$$

so $\mathsf{LR} \neq \mathsf{RL}$ (not commutative). Similarly, $\mathsf{L}^2\mathsf{R} \neq \mathsf{RL}$ (no Thompson-F relation), and $\mathsf{LRL}^{-1} \neq \mathsf{R}^2$ (no standard conjugation relation).

4. Orbit closure reformulation

4.1. The fiber map and backward orbit.

Definition 4.1 (Fiber at 1)**.** For $f = (a, b) \in \mathcal{S}$, the *fiber at* 1 is the unique solution to $f(x) = 1$:

$$\phi(f) \;:=\; f^{-1}(1) \;=\; \frac{1-b}{a} \;\in\; \mathbb{Z}[\tfrac{1}{6}].$$

The *backward orbit of* 1 under $\mathcal{S}$ is $\mathcal{O}^-(1) := \{\phi(f) : f \in \mathcal{S}\} \subset \mathbb{Z}[\frac{1}{6}]$.

Definition 4.2 (Intercept sets)**.** For non-negative integers r, ℓ, let

$$\mathcal{B}(r,\ell) \;:=\; \big\{b_w : w \in \{\mathsf{L},\mathsf{R}\}^*,\ w \text{ has type } (r,\ell)\big\} \;\subset\; \mathbb{Z}[\tfrac{1}{6}].$$

4.2. The equivalence theorem.

Theorem 4.3 (Four equivalent formulations)**.** *The Collatz conjecture is equivalent to each of the following:*

(A) Orbit coverage*: For every $n \in \mathbb{N}$, there exists a word $w \in \{\mathsf{L},\mathsf{R}\}^*$ such that $f_w(n) = 1$.*
(B) Backward orbit*: $\mathcal{O}^-(1) \supseteq \mathbb{N}$.*
(C) Reachability*: For every $n \in \mathbb{N}$, there exist $r, \ell \geq 0$ such that $1 - 3^r n/2^\ell \in \mathcal{B}(r,\ell)$.*
(D) Tree coverage*: The inverse tree $\mathcal{T}$ rooted at 1, with branching rules*

$$n \mapsto 2n \quad \textit{and} \quad n \mapsto \tfrac{n-1}{3}\ \big(\textit{when } n \equiv 1 \pmod 3 \textit{ and } \tfrac{n-1}{3} \textit{ is odd}\big),$$

contains every positive integer as a node.

Proof. **Collatz⇔(A)**: If the Collatz sequence for n terminates at 1, the word $w(n)$ of Remark 2.5 satisfies $f_{w(n)}(n) = 1$ by construction, since each step c_i applies either the even or odd branch of T and the composition carries n to 1. Conversely, if $f_w(n) = 1$ for some $w = c_1 \cdots c_k$, then the sequence $n, f_{c_1}(n), f_{c_1c_2}(n), \ldots, f_w(n) = 1$ is a valid Collatz trajectory from n to 1.

(A)⇔(B): $f_w(n) = 1$ if and only if $n = \phi(f_w) \in \mathcal{O}^-(1)$.

(B)⇔(C): For a type-(r,ℓ) word, $\phi(f_w) = n$ rewrites as $1 - b_w = n \cdot (3^r/2^\ell)$, i.e. $b_w = 1 - 3^r n/2^\ell$. Since $b_w \in \mathcal{B}(r,\ell)$ by definition, this is the condition in (C).

(B)⇔(D): The tree $\mathcal{T}$ is the Cayley graph of $\mathcal{O}^-(1) \cap \mathbb{N}$ under the inverse generators $\phi(\mathsf{L}^{-1})(n) = 2n$ and $\phi(\mathsf{R}^{-1})(n) = (n-1)/3$ (defined when $n \equiv 1 \pmod 3$ and $(n-1)/3$ is odd, so that $T((n-1)/3) = n$ via the odd branch). Coverage of $\mathbb{N}$ is equivalent to $\mathcal{O}^-(1) \supseteq \mathbb{N}$. □

Example 4.4 (Verification for $n = 7$)**.** The Collatz sequence $7 \to 22 \to 11 \to 34 \to 17 \to 52 \to 26 \to 13 \to 40 \to 20 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$ has type $(r,\ell) = (5, 11)$. By (1)–(2),

$$a_{w(7)} = \frac{3^5}{2^{11}} = \frac{243}{2048}, \qquad b_{w(7)} = \frac{347}{2048},$$

and $\frac{243}{2048} \cdot 7 + \frac{347}{2048} = \frac{1701+347}{2048} = \frac{2048}{2048} = 1$. □

4.3. Structure of $\mathcal{B}(r,\ell)$.

Proposition 4.5 (Intercept sets)**.** *For all $r, \ell \geq 0$:*

(i) $|\mathcal{B}(r,\ell)| = \binom{r+\ell}{r}$*: all orderings of a type-(r,ℓ) word produce distinct intercepts.*
(ii) *Every element of $\mathcal{B}(r,\ell)$ has denominator dividing 2^ℓ.*

(iii) $\mathcal{B}(r,0) = \{(3^r-1)/2\}$ *(unique intercept when all* R*'s precede all* L*'s, giving the geometric sum* $1+3+\cdots+3^{r-1} = (3^r-1)/2$*).*
(iv) $\mathcal{B}(0,\ell) = \{0\}$.

Proof. Part (i) is Theorem 3.3. Part (ii) follows from (2): each term has denominator $2^{\ell_i} \le 2^\ell$. Parts (iii)–(iv) are direct computations. □

TABLE 1. Intercept sets $\mathcal{B}(r,\ell)$ for small (r,ℓ).

r	ℓ	$\lvert\mathcal{B}(r,\ell)\rvert = \binom{r+\ell}{r}$	Elements of $\mathcal{B}(r,\ell)$
0	2	1	$\{0\}$
1	1	2	$\{\frac{1}{2},\ 1\}$
1	2	3	$\{\frac{1}{4},\ \frac{1}{2},\ 1\}$
1	3	4	$\{\frac{1}{8},\ \frac{1}{4},\ \frac{1}{2},\ 1\}$
2	2	6	$\{1,\ \frac{5}{4},\ \frac{7}{4},\ 2,\ \frac{5}{2},\ 4\}$
2	3	10	$\{\frac{1}{2},\ \frac{5}{8},\ \frac{7}{8},\ 1,\ \frac{5}{4},\ \frac{11}{8},\ \frac{7}{4},\ 2,\ \frac{5}{2},\ 4\}$

5. Amenability of $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$

Proposition 5.1 (Solvability and amenability)**.** *The group* $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$ *is solvable of derived length* 2*, and therefore amenable.*

Proof. Let $T = \{(1,b) : b \in \mathbb{Z}[\frac{1}{6}]\}$ be the normal subgroup of translations. The quotient $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])/T$ is isomorphic to the multiplicative group $\{2^s3^t : s,t \in \mathbb{Z}\} \cong \mathbb{Z}^2$, which is abelian. Hence $[\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}]), \mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])] \subseteq T$. Since $T \cong (\mathbb{Z}[\frac{1}{6}],+)$ is abelian, $[T,T] = \{0\}$. The derived series $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}]) \supseteq T \supseteq \{e\}$ witnesses solvability of length at most 2. Amenability follows from Day's theorem [4]: abelian groups are amenable, and the class of amenable groups is closed under extensions. □

Remark 5.2 (Thompson's F contrast)**.** Thompson's group F is the group of piecewise-linear homeomorphisms of $[0,1]$ with finitely many breakpoints in $\mathbb{Z}[\frac{1}{2}]$ and slopes in $\{2^k\}$. Whether F is amenable is open; Haagerup–Olesen [5] showed F is not inner amenable. The group $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$ is provably amenable, so proofs requiring amenability of the ambient group are feasible here but not in the Thompson setting.

5.1. Negative drift and density-1 coverage.

Proposition 5.3 (Expected log-decrease per Syracuse step)**.** *Let* n *be odd and let* $k = \nu_2(3n+1)$*. The composed step* RL^k *(odd branch followed by* k *halvings) multiplies* n *by* $3/2^k$*. Taking logarithms, the expected change per Syracuse step is*

$$\mathbf{E}[\log(3/2^k)] = \log 3 - (\log 2)\,\mathbf{E}[k].$$

Under the heuristic that $k = \nu_2(3n+1)$ *is geometrically distributed with* $P(k=j) = 1/2^j$*,* $\mathbf{E}[k] = 2$*, and*

$$\mathbf{E}[\log(3/2^k)] = \log 3 - 2\log 2 = \log(3/4) \approx -0.288.$$

Remark 5.4. Combining the negative drift of Proposition 5.3 with the ergodic theorem for amenable groups (Lindenstrauss [8]) gives logarithmic density 1 for $\mathcal{O}^-(1)$ in $\mathbb{N}$, recovering the quantitative content of Tao's theorem [11] in this algebraic language.

6. The amenability gap and the residual problem

6.1. **What amenability gives and does not give.** Amenability of $\mathrm{Aff}^+(\mathbb{Z}[\frac{1}{6}])$ and negative drift together imply:

$$\frac{1}{\log N} \sum_{\substack{n \le N \\ n \notin \mathcal{O}^-(1)}} \frac{1}{n} \longrightarrow 0.$$

This is a density statement. It does *not* imply full coverage, because a set of logarithmic density 1 may still omit finitely (or even infinitely many sparse) integers. We call this the *amenability gap*.

6.2. **The open problem.**

Open Question 6.1 (Subset-sum reachability). For every $n \in \mathbb{N}$, do there exist $r, \ell \ge 0$ such that

$$1 - \frac{3^r n}{2^\ell} \in \mathcal{B}(r, \ell) \ ?$$

An affirmative answer to Question 6.1 is equivalent to the Collatz conjecture (Theorem 4.3(C)).

Remark 6.2 (Feasibility of the target)**.** By Euler's theorem (since $\gcd(3, 2^\ell) = 1$), for each n and each ℓ there exists r with $3^r n \equiv 1 \pmod{2^\ell}$, making $1 - 3^r n/2^\ell \in \mathbb{Z}[\frac{1}{6}]$. So the condition is non-vacuous: for every n and ℓ there is at least one (r, ℓ) for which the target lands in $\mathbb{Z}[\frac{1}{6}]$. The question is whether it lands in the smaller set $\mathcal{B}(r, \ell)$.

Remark 6.3 (Density of targets)**.** Since $\log 3/\log 2$ is irrational, the set $\{3^r/2^\ell : r, \ell \ge 0\}$ is dense in $\mathbb{R}_{>0}$. For fixed n, the targets $1 - 3^r n/2^\ell$ are therefore dense in $\mathbb{R}$ as (r, ℓ) varies. Combined with the fact that $|\mathcal{B}(r, \ell)| = \binom{r+\ell}{r} \to \infty$, this makes the reachability condition plausible but does not prove it.

7. The $\mathrm{Im}(\tau)$ dynamics and the polylogarithm

A parallel line of investigation, predating the affine embedding, attaches a complex parameter $\tau(n)$ to each iterate and studies its dynamics. The results here are self-contained and complement the affine picture.

Definition 7.1 (τ-parameter)**.** For $n \in \mathbb{N}$ let $a_0 = \nu_2(n)$ and $a_1 = \nu_3(n)$ (the 2- and 3-adic valuations). Define

$$\tau(n) = \frac{a_1 + i\,\nu_2(3n+1)}{a_0 + 1} \in \mathbb{C}.$$

Proposition 7.2 (Imaginary axis dominance)**.** *For almost all iterates of the Collatz map, $a_1 = 0$ and hence $\mathrm{Re}(\tau) = 0$. The sequence $\tau(n), \tau(T(n)), \ldots$ lives almost entirely on the positive imaginary axis, and the dynamics reduce to the real sequence $y_k := \mathrm{Im}(\tau(T^k(n))) \in \mathbb{R}_{>0}$.*

Proposition 7.3 (Rational transition rules)**.** *The sequence (y_k) obeys:*

$$\textit{Even step } (n \to n/2): \quad y_{k+1} = y_k \cdot \frac{a_0^2 - 1}{a_0^2} = y_k \cdot \left(1 - a_0^{-2}\right),$$

$$\textit{Odd step } (n \to 3n+1): \quad y_{k+1} = y_k \cdot \frac{1}{\nu_2(3n+1) + 1}.$$

Every sequence terminates at $y = \frac{1}{2}$ (corresponding to $n = 2$).

Proposition 7.4 ($\pm S$ symmetry and the polylogarithm)**.** *Define $S = \sum_{k=1}^{\infty} k(\log k)/2^k \approx 0.5078$. Then*

$$\mathbf{E}[\Delta \log y \mid \textit{odd step}] = +S, \qquad \mathbf{E}[\Delta \log y \mid \textit{even step}] = -S.$$

The constant S equals the negative of the derivative of the polylogarithm at $s = 0$:

$$S \;=\; -\frac{d}{ds}\,\mathrm{Li}_s\!\left(\tfrac{1}{2}\right)\Big|_{s=0}.$$

Proof sketch. Under the heuristic that $\nu_2(3n+1)$ is geometrically distributed, the expected log-change per odd step is $\mathbf{E}[\log(1/(\nu_2+1))] = -S$ with sign corrected by the direction of y. The polylogarithm identity follows from the series expansion $\mathrm{Li}_s(x) = \sum_{k=1}^{\infty} x^k k^{-s}$ differentiated at $s = 0$. □

Remark 7.5. The $\pm S$ symmetry implies that $\log y_k$ is a zero-mean random walk (conditionally on equal step probabilities). This is consistent with the Collatz conjecture but does not prove it: the steps are deterministic (not independent), $E[\text{ratio}] > 1$ for the $\mathrm{Im}(\tau)$ process itself, and the terminal condition $y = \frac{1}{2}$ is not equivalent to $n = 1$.

8. Directions for further work

1. **Explicit fiber construction.** For each $n \in \mathbb{N}$ and each $\ell \geq 1$, there exists r with $3^r n \equiv 1 \pmod{2^\ell}$ (by Euler's theorem). A proof of the conjecture via (C) would need to show that, for such (r, ℓ), the target $b^* := 1 - 3^r n/2^\ell$ lies in $\mathcal{B}(r, \ell)$. A natural approach is to show that b^* is always expressible as an ordered partial sum of the form (2).
2. **Minimum gap in $\mathcal{B}(r, \ell)$.** Quantifying $\min_{b \neq b' \in \mathcal{B}(r,\ell)} |b - b'|$ would bound how large (r, ℓ) must be for the target b^* to have any chance of lying in $\mathcal{B}(r, \ell)$. Since $|\mathcal{B}(r, \ell)| = \binom{r+\ell}{r}$ and all elements have denominator $\leq 2^\ell$, the minimum gap is at least $2^{-\ell}$.
3. **The $(2, 3)$-Thompson group $G_{(2,3)}$.** The group of piecewise-linear homeomorphisms of $[0, 1]$ with breakpoints in $\mathbb{Z}[\frac{1}{6}]$ and slopes in $\{2^s 3^t\}$ is the natural rigid ambient group; it contains $\mathcal{S}$ as an infinite-type element. Establishing that its derived subgroup is simple and computing its abelianisation (conjectured $\mathbb{Z}^2$, using the Stein–Brown framework [10]) would be the first new result in this direction.
4. **C*-simplicity.** If $G_{(2,3)}$ is C*-simple, any action on a compact space has a unique stationary measure, strongly constraining the orbit structure. This could close the amenability gap by ruling out exceptional orbits.
5. **Furstenberg's $\times 2$, $\times 3$ connection.** The slope group $\{2^s 3^t\}$ is the same multiplicative structure appearing in Furstenberg's conjecture on simultaneously $\times 2$- and $\times 3$-invariant measures on $\mathbb{R}/\mathbb{Z}$. Recent progress by Shmerkin [9] and Host [6] may be relevant to understanding the distribution of the semigroup action.

Acknowledgements

These results emerged from a computational investigation beginning with a Godel-encoding approach (Section 7), proceeding through $\mathrm{Im}(\tau)$ dynamics and the $\pm S$ symmetry, and culminating in the affine embedding of Sections 3–6. All algebraic computations were verified exactly using rational arithmetic over $\mathbb{Z}[\frac{1}{6}]$.

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